DHANUSH DABBIRU

Artificial Intelligence Engineer / ML Engineer

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EDUCATION

Master of Science in Computer Science, George Mason University **Aug 2023 — May 2025**

GPA: 3.47/4.00 Concentration: Machine Learning

Bachelor of Science in Computer Science, GITAM University **Jun 2019 — Apr 2023**

CGPA: 8.44/10

SKILLS

Languages	Python, Java, C++, C
AI/ML	PyTorch, HuggingFace, LangChain, Ollama, FAISS, EasyOCR
LLM	LLaMA (3.2, 4), Gemma, Qwen, GPT-4, RAG, Prompt Engineering
Frameworks	Flask, Flutter, Unity, Unreal Engine
Front-End	HTML, CSS, JavaScript, Dart
Cloud	AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure
Tools	Git, Docker, Arduino IDE, Android Studio
Hardware	Raspberry Pi, NodeMCU, Arduino

EXPERIENCE

Areaprobe, Washington, DC **Artificial Intelligence Engineer** **Sept 2024 — Present**

- Designed and deployed **Resident's View (Resi)** with **LLaMA3.2-Vision:11B**; contextual scene descriptions and 24/7 alerts to 100+ endpoints.
- Integrated **Kokoro TTS**; improved pilot engagement by 40%.
- Benchmarked **VLMs** (**LLaMA4:16x17B**, **Gemma3:12B**, OpenAI); +25% description accuracy.
- Built scalable **RAG pipeline** (FAISS + all-MiniLM-L6-v2) over 1M+ docs, sub-200ms latency.
- Applied advanced **prompt engineering** (few-shot, negatives, CoT) cutting hallucinations by 30%.
- Tuned **decoding parameters** (temperature, top-p, top-k) for +20% coherence; shipped secure **Flask APIs** with real-time streaming.

Resi Suite: *Secure* (YOLOv12n, -18% false positives), *Detect* (YOLOv11n COCO, 80+ classes @30+ FPS), *Read* (YOLOv12n+11n+EasyOCR, 92% plate accuracy), *Recognize* (InsightFace, <100ms).

Kairos, Remote **AI Foundational Engineer** **Jun 2025 — Present**

- Developed **Kairos** assistant integrating **LLaMA4:Scout**, **VOSK STT**, **Chatterbox TTS**.
- Delivered **hands-free interface** with custom zero-shot voices; boosted realism and engagement.
- Designed modular **Flask deployment** enabling rapid third-party **API integration**.

ACHIEVEMENTS

- Microsoft: Azure AI-900; AWS: CLF-C02, AIF-C02; 100+ LeetCode (Top 20%): Portfolio
- Research: IoT Enabled Aqua Pond Monitoring System (IEEE); Wi-Fi Vehicle Control (Springer)

PROJECTS

- **Smart Home Lighting Automation (Feb–Mar 2025):** Built **agentic workflow** in **n8n** with **ChatGPT** to control **Philips Hue** bulbs; **automated 100% of requests** with $<$1s **latency**. **Tech:** n8n, ChatGPT, Philips Hue API
- **Personal Finance Assistant (Apr–May 2025):** Developed agent on **Ollama Gemma3:4B (4-bit)** for **categorization**, **anomaly detection**, and **insight generation**; reduced bookkeeping by **90%** with **95% accuracy**. **Tech:** Ollama, Python, Google Sheets API
- **Email Triage Assistant (Feb–Mar 2025):** Implemented **triage assistant** with **HuggingFace + Qwen3:4B** to **prioritize**, **draft replies**, and **auto-organize inbox**; cut handling time by **70%**. **Tech:** HuggingFace, Qwen3:4B, Gmail API, Python